The broadnose sevengill shark, *Notorynchus cepedianus*, an apex predator: population structure and growth estimates

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Abstract

Large sharks are globally distributed predators that contribute to the stability and functioning of marine ecosystems. Accurate estimates of their population size structure and growth dynamics are essential for effective conservation and management in the face of ongoing unsustainable fishing pressure. This is particularly critical for the broadnose sevengill shark (*Notorynchus cepedianus*) on the Southwestern Atlantic continental shelf, where this highly mobile species has experienced significant population declines. Its wide-ranging movements, combined with the broad extent of the continental shelf in the region, pose challenges for accurately characterizing its demog-

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raphy. In this context, long-term monitoring studies emerge as a powerful approach for generating critical biological information on the size-structure and growth of demersal migratory species such as N. cepedianus, thereby supporting robust population assessments and evidence-based management recommendations. To this end, we integrated over 15 years of tag–recapture data collected in coastal waters of the Southwestern Atlantic ($\sim$34.6°S to $\sim$49.4°S), including records of sex and total length, and applied three nonlinear growth models to estimate growth dynamics. The results reveal distinct regional variations in size structure across the latitudinal range. The highest density of large specimens were recorded in the central region (41.2°S $<$ latitude $\leq$44.0°S), whereas the smallest individuals were predominantly captured in the north ($\leq$41.2°S). Age–growth curves suggest that female recruit to their migratory phase at 2 – 7 years and reach maturity at 8 – 15 years. These findings underscore the potential of long-term tagging programs to support effective management and regional conservation strategies and actions.

Keywords: Southwestern Atlantic, Hexanchidae, Gompertz model, Tag-recapture, Ontogenetic migration

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1. Introduction

The broadnose sevengill shark, *Notorynchus cepedianus*, is a large predator distributed across continental and insular shelves - including bays, estuaries, and nearshore waters in tropical and temperate seas (Compagno, 1984; Compagno et al., 1989; Last and Stevens, 1994). Understanding its population structure and growth patterns is essential for assessing ecosystem health and informing conservation strategies at both regional and global scales. This is especially critical as *N. cepedianus* has faced significant population reductions documented at both scales (Barbini et al., 2015; Irigoyen and Trobbiani, 2016; Finucci et al., 2020), mirroring the widespread declines observed across various chondrichthyan taxa (Dulvy et al., 2024).

In the Southwestern Atlantic (SWA), its range extends from 25° to 52°S (Sadowsky, 1969; Guzmán and Campodónico, 1976), with its main distribution along the Uruguayan and Argentine coasts (33 – 50°S). *Notorynchus cepedianus* is considered particularly relevant within the SWA ecosystem (De Wysiecki et al., 2020), alongside the killer whale (Orcinus orca), it is one of the few apex predators in the region (Funes et al., 2024). A theoretical migration model for the species has described two principal displacement patterns in the SWA. The northward movement involves individuals migrating from Patagonian waters (40° – 47°S) to the Río de la Plata (33° – 37°S) on autumn, whereas the southward movement represents a return migration along coastal waters, reaching southern latitudes in the summer (De Wysiecki et al., 2020). Additionally, several studies conducted at discrete coastal aggregation sites have documented seasonal patterns of occurrence (Cedrola et al., 2009; De Wysiecki et al., 2018; Irigoyen et al., 2019), and neonates are frequently recorded during late spring and summer in northern areas (36.5° – 40°S; Lucifora et al., 2005; Jaureguizar et al., 2022, 2023).

Population structure and growth estimates, two critical parameters for conservation management, are commonly derived from length-weight relationships and length-at-age data (Erzini, 1994; Froese, 2006; Compaire et al., 2024). However, implementing such studies for large, highly mobile species like sharks is inherently challenging. Their ability to move across vast distances, coupled with sporadic captures in seasonal or multi-species fisheries, often results in fragmented data that complicates population assessment. Additionally, in chondrichthyans, classical age-determination techniques are often unreliable because of their poorly calcified cartilaginous skeletons, making the estimation of accurate ages particularly challenging (Matta et al.,

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2017). Consequently, long term tag-recapture experiments have emerged as a successful technique to study growth and movements of mobile marine predators, as they integrate efforts from scientists, recreational fishers, and the fishing industry (Simpfendorfer, 2000; Pilling et al., 2007). This study aims to provide novel insights into the population structure and growth dynamics of *N. cepedianus* in the SWA using data from wild populations for the first time. Integrating over 15 years of data from two tagging programs, this work provides essential life-history information for the fisheries management of the species and enables the estimation of the generation length, which constitutes fundamental baseline information for conservation status assessments (Mace et al., 2008).

2. Material and Methods

2.1. Study site and data collection

Sampling was conducted in coastal waters of the Southwestern Atlantic (SWA) (∼34.6°S to ∼49.4°S) between January 2011 and January 2026, under two main tagging programs. The first program has been conducted since 2011 by the Fish Ecology Laboratory of CESIMAR as part of a series of studies carried out in Golfo Nuevo (∼42.7°S; Irigoyen et al., 2015), Caleta Valdés (∼42.4°S; Irigoyen et al., 2018), Caleta Malaspina (∼45.1°S; Irigoyen et al., 2019), and most recently in the Puerto Deseado Estuary (∼47.8°S). The latter in collaboration with the research and shark conservation project of the NGO *Por el Mar* (Figure 1). The fishing effort for tagging generally occurred throughout all seasons of the year, with additional fishing trips conducted during periods of peak abundance at each site (Cedrola et al., 2009; Irigoyen et al., 2018, 2019). The second program, *Conservar Tiburones en Argentina*, was launched in 2013, to engage shark anglers in shark conservation. Since 2010 anglers (n = 265) were trained and engaged in conventional tagging duties, best fishing practices and shark conservation issues with 126 anglers still active today along > 2,500 km of coastline from San Clemente del Tuyú (∼36°S) to Península de San Julián (∼49°S), covering 41 fishing sites. All captured fish were tagged using Dart-type tags (models FT-1-94 or PDS; Floy Tag and Hallprint Inc.) and basic biological parameters were recorded, including sex and total length (TL) to the nearest cm, using a fiberglass tape.

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2.2. Length and growth analyses

To evaluate differences among specimens collected across a wide latitudinal range, coastal waters were divided latitudinally into three regions based on sampling effort: north ($\leq$ 41.2°S), central (41.2°S $<$ latitude $\leq$ 44.0°S), and south ($>$ 44.0°S). Differences in TL between sexes and among regions were evaluated using the non-parametric Kruskal–Wallis test, as the data did not meet the assumptions of normality and homoscedasticity. Post-hoc pairwise comparisons among regions were performed using Dunn's test with Bonferroni correction for multiple comparisons. Kernel density estimation curves were overlaid on histograms to provide a smoothed representation of total length distributions, reducing sensitivity to bin width and facilitating visual comparison among regions.

Three non-linear growth models (Gompertz (1825), Robertson (1923), von Bertalanffy growth function - VBGF - parameterized following Fabens (1965) were fit to the tag release-recapture data for *N. cepedianus*. Since absolute ages were unknown, we utilized the interval forms of these equations, which relate the length at release (L_1) and recapture (L_2) over a time interval (Δt). Based on empirical size-at-birth and maximum size estimates reported for the study region, the length at birth (L_0) and asymptotic (length L_∞) were fixed at 40 cm (Jaureguizar et al., 2022), and 280 cm (Milessi, 2008), respectively. Fixing L_∞ reduced parameter correlation and improved model stability when fitting tag–recapture growth models. The Francis (1988) parametrization was evaluated but yielded identical likelihoods and growth trajectories to the Fabens formulation under fixed L_∞, and was therefore not treated as an independent model. In all models, K denotes the growth coefficient. However, it is important to note that despite K representing a rate parameter in all models, its mathematical definition and biological interpretation depend on the functional form. In the VBGF - Fabens (1965), K determines the rate at which length approaches the asymptotic length L_∞ (Pauly, 1980). In contrast, in sigmoidal models such as the Gompertz (1825) and logistic functions Robertson (1923), K describes the exponential decay of the relative growth rate, and the resulting growth trajectories are characterized by the presence of an inflection point (Katsanevakis and Maravelias, 2008). The models are described as follows:

Gompertz (1825): A model where the growth rate decreases exponentially with size, suitable for non-symmetric sigmoidal growth:

$$L_2 = L_\infty \cdot e^{\ln\left(\frac{L_1}{L_\infty}\right)e^{-K\Delta t}} \tag{1}$$

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where L_1 and L_2 represent the total length at release and recapture, respectively, and Δt is the time interval at liberty.

Fabens (1965): A non-linear method that fits data directly to the VBGF by minimizing the residuals between observed and predicted recapture lengths:

$$L_2 = L_\infty - (L_\infty - L_1)e^{-K\Delta t} \tag{2}$$

Robertson (1923): Also known as the Logistic model, it describes a sigmoidal growth curve symmetric around its inflection point:

$$L_2 = \frac{L_\infty}{1 + \left(\frac{L_\infty}{L_1} - 1\right) e^{-K\Delta t}} \tag{3}$$

Data for the growth analysis was restricted to individuals with reliable total length measurements at both release and recapture. To ensure the integrity of the dataset and minimize the influence of potential measurement errors, the analyses were performed on specimens exhibiting annual growth rates up to 40 cm year^{-1} (Van Dykhuizen and Mollet, 1992). Because growth rates in elasmobranchs can be sex-dependent (Braccini et al., 2010; Liu et al., 2021; Simpfendorfer, 2000), we evaluated the relationship between length increment and time at liberty for males and females separately prior to model fitting. Residual diagnostics were conducted only for the non-linear models fitted by maximum likelihood assuming homoscedastic Gaussian errors on recapture length. These diagnostics included tests for normality (Shapiro–Wilk), length-dependent residual structure (Spearman rank correlation between residuals and initial length), and non-random residual structure (Wald–Wolfowitz runs test). The performance among models was evaluated using the bias-corrected Akaike's Information Criterion (AICc) (Burnham and Anderson, 2002). Because the absolute value of the AICc by itself is not informative, we computed the differences in AICc from the minimum AICc value to assess relative model performance as ΔAICc = $AICc_i$ - $AICc_{min}$. The ΔAICc values allow for a quick comparison and ranking of candidate models. Models with ΔAICc < 2 have statistical support (Burnham and Anderson, 2002). For visualization purposes, age-based growth curves were reconstructed by anchoring all models at size at birth mentioned above.

2.3. Data processing and software

Statistical analyses were performed using Python (version 3.12.2; Python Software Foundation, https://www.python.org). Non-parametric compar-

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isons of TL were conducted using the SciPy library (v1.16.1; Virtanen et al., 2020). When significant differences were detected, post hoc pairwise comparisons were performed using Dunn's test as implemented in the scikit-posthocs package (v0.11.4; Terpilowski, 2019). Additional statistical diagnostics and data exploration were supported by Statsmodels (v0.14.5; Seabold and Perktold, 2010).

3. Results

Out of 1,133 *N. cepedianus* specimens identified, total length (TL) was recorded for 1,109. Individuals ranged from 45 to 265 cm ($\bar{x} \pm \mathrm{SD} = 180.5 \pm 44.1$ cm; Figure 2A), with females being significantly larger than males (185.1 ± 46.5 cm vs. 169.8 ± 33.8 cm; KW test: $H(1) = 42.81$, $p < 0.001$). Significant regional differences were also observed ($H(2) = 414.47$, $p < 0.001$); specimens from the central region (208.56 ± 27.65, n = 511) were larger than those from the south (166.23 ± 27.51, n = 254) and north (149.45 ± 47.81, n = 344) (Dunn test, $p < 0.05$), while the latter two showed no differences (Dunn test, $p > 0.05$). Intra-regional analyses revealed that females were significantly larger than males in the central (KW test: $H(1) = 106.97$, $p < 0.001$) and southern (KW test: $H(1) = 9.30$, $p = 0.002$) regions, whereas no significant difference between sexes was detected in the northern region (KW test: $H(1) = 1.63$, $p = 0.202$). A detailed summary of abundance and length statistics by region and sex is provided in Table 1, while their respective length frequency distributions are illustrated in Figure 2B and 2C).

From 71 tag–recaptured specimens, 50 individuals remained after applying annual growth-rate filtering criteria. A significant positive relationship between length increment and time at liberty was detected only in females ($p < 0.001$, n = 37), whereas no such relationship was observed in males ($p = 0.433$, n = 13). Consequently, growth model fitting and comparison were conducted exclusively using female specimens. Tag-recapture selected females were at liberty between 12 and 1488 days, and their size ranged from 133 to 261 cm. Estimates of growth coefficient (K) ranged from 0.1220 to 0.1688 yr^{-1} among candidate models. Residual diagnostics yield that all models provided a robust fit to the data, satisfying the assumptions of normality (Shapiro-Wilk, $p > 0.05$), lack of length-dependency (Spearman, $p > 0.05$), and randomness in residual distribution (Runs test, $p > 0.05$). The three models received substantial support ($\Delta\mathrm{AICc} < 2$) for estimating the growth coefficient in *N. cepedianus* females (Table 2).

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The growth curves generated by the Fabens (VBGF), Gompertz, and Robertson models (Figure 3) illustrate consistent asymptotic behavior toward the fixed value of L_∞, but with marked differences in growth trajectories during early and intermediate life stages. The Fabens (VBGF) model predicts the fastest early growth, reaching larger sizes at younger ages compared with the other two models. In contrast, the Robertson model exhibits a more gradual increase in length, particularly for small individuals, resulting in lower predicted sizes at intermediate ages. Lastly, the Gompertz model displays an intermediate pattern, with growth rates lower than those of the Fabens (VBGF) model at early ages but higher than the Robertson model.

4. Discussion

This study assessed the population structure and growth rates of *N. cepedianus* in the coastal waters of the Southwestern Atlantic (SWA) using 15 years of tag release-recapture data. Our findings reveal distinct latitudinal variations in size structure; the highest density of large specimens were recorded in the central region, followed by those from the southern waters, while the distribution peaked at smaller sizes for individuals captured in the northern region. This spatial pattern in size structure across regions corroborates previous reports identifying northern Argentina and Uruguay as the species' primary parturition and neonate nursery areas (Silveira et al., 2018; De Wysiecki et al., 2020; Jaureguizar et al., 2023), as highlighted by the presence of smaller individuals (45 to 100 cm TL) in the northern region. The size structure of individuals from the central zone indicates that organisms are significantly larger and, consequently, include a high proportion of adults of both sexes. This demographic distinction is largely driven by data from Caleta Valdés, which is the only known aggregation site for both mating and feeding purposes in the SWA (Irigoyen et al., 2018; Funes et al., 2024). Hence, the observed size-structure pattern in the central zone may be biased by the concentration of samples obtained from a reproductive area of the species.

The length range analyzed to assess population size structure and growth rates in the SWA was similar to that reported in other regions for wild populations (52 – 284 cm in southern Australia; Braccini et al., 2010; 61 – 286 cm in southern Africa; Engelbrecht et al., 2020) and captive specimens (48 – 240 cm at Melbourne Aquarium; Braccini et al., 2010; 65 – 188 cm at Monterey Bay Aquarium; Van Dykhuizen and Mollet, 1992). To accu-

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rately translate the observed length distributions into growth estimates, we employed a multi-model approach as recommended for elasmobranchs (Liu et al., 2021). We compared three non-linear candidate models commonly utilized in shark growth studies: the VBGF via Fabens (1965), Gompertz (1825), and Logistic (Robertson, 1923) models (Van Dykhuizen and Mollet, 1992; Cailliet and Goldman, 2004; Cailliet et al., 2006). All models exhibited adequate fit to the data and revealed a pattern of rapid early growth followed by progressive deceleration as individuals approach the asymptotic length ($L_\infty = 280$ cm). Given that all three models received strong statistical support (ΔAICc < 2), our results indicate a lack of clear support for a single growth function describing the growth trajectory of *N. cepedianus* females in the SWA. The inability to discriminate among alternative models reflects a common limitation of tag–recapture growth studies, in which tagging data contain no unambiguous information on absolute age (Francis, 1988), and therefore potential age-dependent variation in growth cannot be explicitly modelled. In the absence of such information, our data do not allow us to determine whether early growth in *N. cepedianus* females follows the strictly trajectory of the VBGF or the sigmoidal patterns predicted by the Gompertz and Logistic models. Despite this model uncertainty, all candidate models predicted comparable annual length increments across the observed size range of individuals in the dataset, where the confidence intervals of the growth trajectories broadly overlap. Although early growth trajectories differ among model formulations, the overall magnitude of growth inferred for *N. cepedianus* females appears robust to model choice. This consistency increases confidence in the overall magnitude of growth rates inferred for *N. cepedianus* females.

The estimated growth coefficient for females in this study derived from tag-recapture data based on the VBGM parametrization following Fabens falls within the range of previous estimates reported for captive females, which vary from K = 0.122 to 0.235 (Braccini et al., 2010) and reach up to K = 0.29 (Van Dykhuizen and Mollet, 1992). However, this estimate remains higher than those reported for other wild female populations (K = 0.072 to 0.089; Braccini et al., 2010). In contrast, it is considerably lower than the rapid growth rates observed for wild young-of-the-year females in the SWA, where growth coefficients ranged from 0.0689 to 0.3741 for the VBGF, 0.8508 to 1.5717 for the Gompertz model, and 1.6374 to 2.2823 for the Logistic model (Jaureguizar et al., 2022). Based on growth curves reconstructed in this study, *N. cepedianus* females would recruit to its migratory phase -

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moving into the central and southern regions - at around 100 cm TL, which corresponds to an estimated age between 2 and 7 years. Following this age-growth reconstruction and the female maturity length previously reported for the SWA (190 cm TL; Irigoyen et al., 2018), females in this region would reach sexual maturity at between 8 and 15 years of age. The total of males tagged and recaptured was insufficient to provide a robust estimation of growth parameters for this sex.

The number of males captured was substantially lower than that of females, representing approximately a 1:3 ratio. Three non-exclusive explanations may account for this pattern: (1) coastal areas, where fishing effort was concentrated, may be used less frequently by males; (2) males may experience higher mortality; or (3) males may show lower selectivity for the bait or hook-based fishing gear. While empirical data to support these hypotheses are currently lacking, the first explanation – spatial segregation– appears to be the most plausible (Irigoyen, pers. obs.) and could be further evaluated through satellite telemetry by tracking individuals of both sexes. If this hypothesis were correct, males would use the Argentine shelf for a significantly longer period of time than females, remaining outside the spatial coverage of the coastal sampling effort.

5. Conclusions

Our results align with previous studies and provide new insights into the distribution of *Notorynchus cepedianus* in the Southwestern Atlantic across its ontogeny. Younger specimens primarily inhabit northern nursery grounds (Jaureguizar et al., 2023), whereas older individuals tend to occupy the central and southern regions. The age-growth reconstruction curves indicate that females recruit to meridional regions at approximately 100 cm TL, which corresponds to an estimated age of 2 – 7 years, depending on the growth model used. Considering its very rapid growth during the first year of life reported for this species in the region (Jaureguizar et al., 2022), the estimate of 2 years derived from the VBGM parametrization following Fabens appears to be the most plausible, at least for this early life stage. Following this ontogenetic shift, and based on the maturity length reported for the region (Irigoyen et al., 2018), females would subsequently reach sexual maturity between 8 and 15 years of age. Given the high mobility and wide distribution of this species, the findings presented here reinforce the suitability of long-term tagging programs for providing critical biological information on population size

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structure, and growth. Such data are essential for the effective management and regional conservation of *N. cepedianus* in the Southwestern Atlantic, as well as for strengthening monitoring frameworks and future population assessments. In particular, the biological parameters estimated in this study provide key inputs for detecting temporal changes in demographic structure and for estimating generation length based on data from wild populations for the first time (Finucci et al., 2020). This constitutes critical baseline information for IUCN Red List conservation status assessments and adaptive management strategies and actions across the region.

Author contributions

Jesus C. Compaire: Methodology, Software, Formal analysis, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization. **Gastón Trobbiani:** Resources, Conceptualization, Writing - Review & Editing, Visualization. **Federico Mas:** Resources, Writing - Review & Editing. **Juan Martín Cuevas:** Resources, Conceptualization, Writing - Review & Editing. **Carolina Pantano:** Resources, Conceptualization, Data Curation, Writing - Review & Editing. **Martín Laporta:** Resources, Writing - Review & Editing. **Lucas Albornoz:** Resources, Writing - Review & Editing. **Cristian Lagger:** Resources, Writing - Review & Editing. **Alejo Irigoyen:** Conceptualization, Methodology, Investigation, Resources, Data Curation, Writing - Original Draft, Writing - Review & Editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used AI in order to improve readability and language. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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Data available on request from the corresponding author.

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Figures

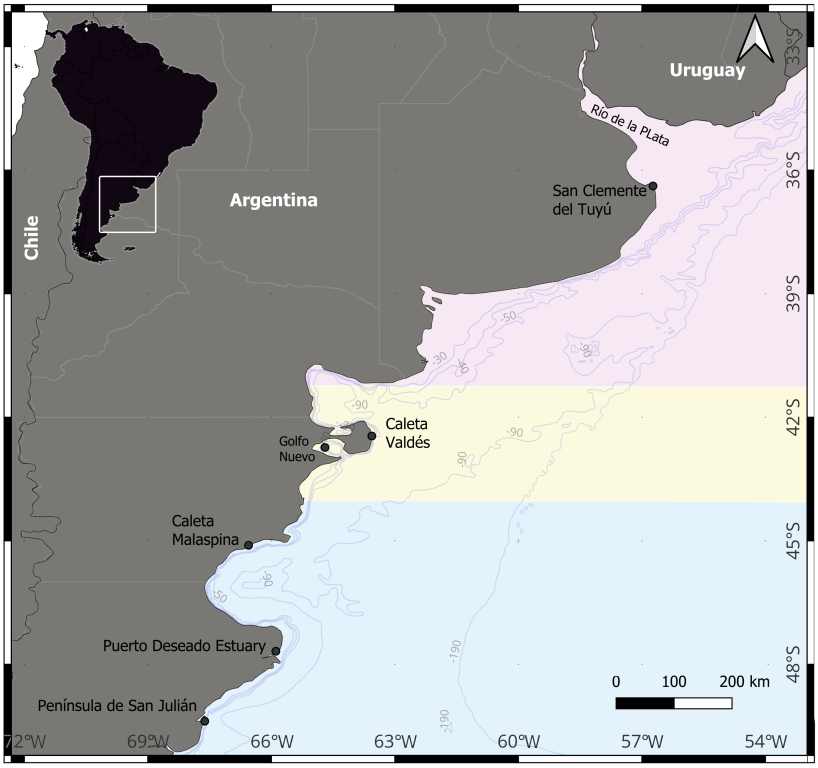

Figure 1: Study area for *Notorynchus cepedianus* in the southwestern Atlantic Ocean (∼34.6°S to ∼49.4°S). The pale colored shading over the sea indicates regional divisions defined according to the latitudinal thresholds applied in the analyses: north (pink, ≤41.2°S), central (yellow, 41.2°S < latitude ≤ 44.0°S) and south (blue, latitude > 44.0°S). Bathymetric contours (isobaths) are depicted to illustrate the coastal shelf relief.

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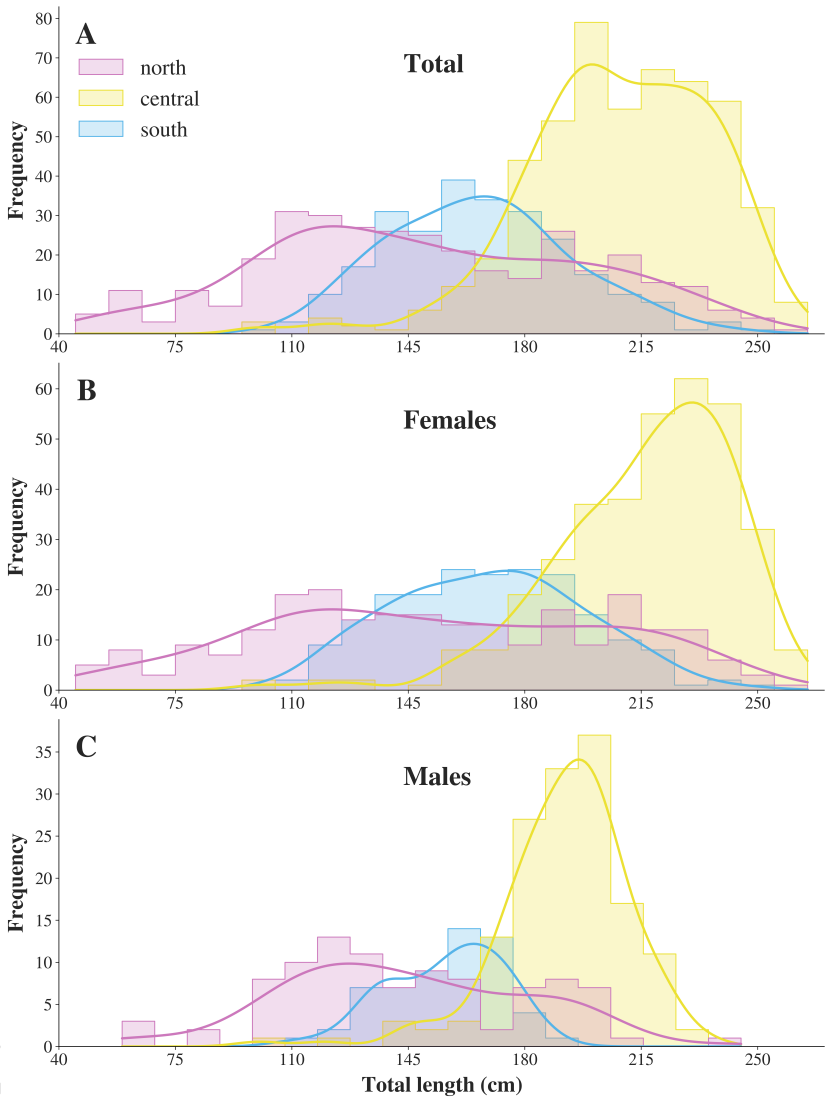

Figure 2: Total length (TL) distribution of *Notorynchus cepedianus* for the entire sample (A) and by sex (B: females, C: males) for each region. Regions were defined based on latitude as follows: north ($\leq$ 41.2°S), central (41.2°S $<$ latitude $\leq$ 44.0°S), and south ($>$ 44.0°S). Histograms are presented with 10 cm bins, and kernel density estimation curves are overlaid on the histograms to illustrate the smoothed distribution of total length for each region.

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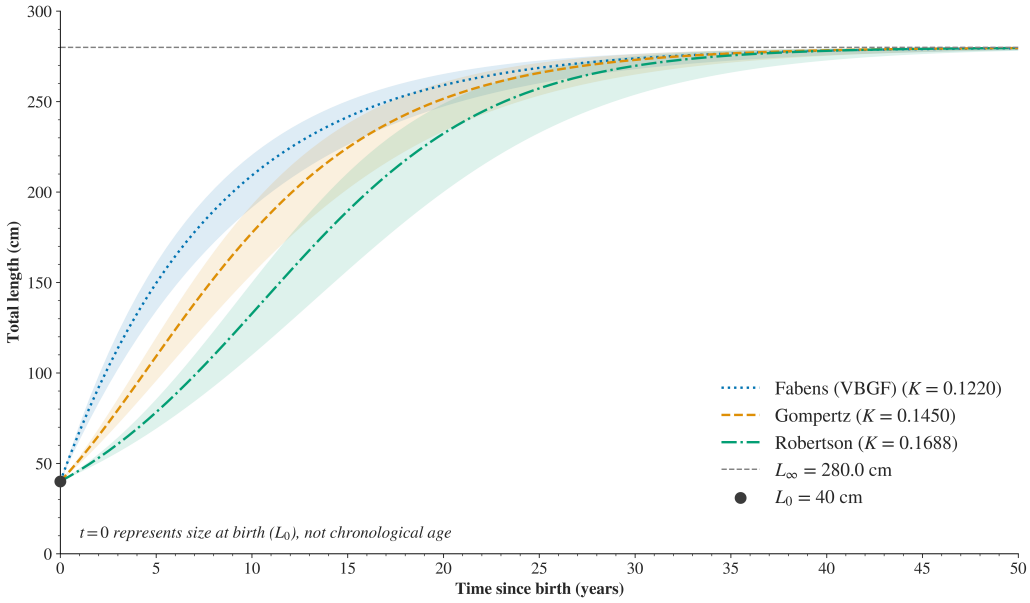

Figure 3: Growth curves for tag – recaptured *Notorynchus cepedianus* females estimated using three non-linear models: the von Bertalanffy growth function (VBGF) parameterized using the Fabens method (dotted blue line), Gompertz (dashed orange line), and Robertson (dashed-dotted green line). Shaded areas represent the 95% confidence interval for each curve. The length at birth (L_0) is depicted by a black dot ($t = 0$), and the fixed asymptotic length (L_∞) by a grey dashed line. Short horizontal lines along the y-axis indicate the observed range of total lengths used to estimate growth.

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Tables

Table 1: Sample size (n), total length (TL) range, and mean TL ($\pm$ SD) of *Notorynchus cepedianus* disaggregated by region and sex in the Southwestern Atlantic.

Region	Sex	n	TL range (cm)	TL mean ± SD (cm)
North	Female	240	45 – 260	152.20 ± 51.69
	Male	97	59 – 245	144.06 ± 36.52
Central	Female	357	100 – 265	216.06 ± 26.87
	Male	150	100 – 235	191.78 ± 19.33
South	Female	195	103 – 249	168.86 ± 28.98
	Male	58	110 – 188	156.17 ± 17.11

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Table 2: Estimates of the growth coefficient (K) and corresponding 95% confidence intervals, Akaike's Information Criterion differences (ΔAICc) for comparison of candidate non-linear growth models fitted by maximum likelihood estimation, and residual diagnostics for tag–recapture growth models of *Notorynchus cepedianus*. Residual diagnostics include the Shapiro–Wilk test for normality (W), Spearman rank correlation test for length-dependence (ρ), and the runs test for randomness (Z); reported values are p-values.

Models	K (95% CI)	ΔAICc	W	ρ	Z
Fabens (VBGF)	0.1220 (0.0996 – 0.1395)	1.0271	0.7979	0.5350	0.0671
Gompertz	0.1450 (0.1187 – 0.1652)	0	0.6272	0.9439	0.0710
Robertson	0.1688 (0.1354 – 0.1934)	1.1901	0.3344	0.6050	0.0710

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